■# BDA501 Individual Assignment 3 Report

Land-Cover Classification at Scale with Spark MLlib.

1. Environment and data

• Platform: Windows local Spark / PySpark MLlib, local HDFS mirror `hdfs_mirror/bigearthnet`.

• Dataset source: BigEarthNet HDF5 partial data from Assignment 2.

• Rows used: 52,607 patches across 4 countries.

• Official split counts: {'train': 25291, 'validation': 13884, 'test': 13432}.

• No random split or random-fold cross-validation was used; train/validation/test columns are preserved.

• AI assistance: OpenAI Codex helped implement and verify scripts; all code/results are reproducible from this repo.

2. Step 1: Richer features

Features v2 saved to `file:///C:/Users/DELL/Documents/Bigdata-Lab/Lab2-Bigdata/hdfs_mirror/bigearthnet/features_v2`.

Final feature count: 28.

The new 20 m bands B05, B06, B07, B8A, B11 and B12 are processed on their own 60 x 60 grid. NDMI, NDBI, and NDVI p10/p50/p90 are computed per patch.

Validation: 0 feature errors; null counts and NaN counts are all zero; index range failures: {'ndvi': 0, 'ndwi': 0, 'ndmi': 0, 'ndbi': 0, 'ndvi_p10': 0, 'ndvi_p50': 0, 'ndvi_p90': 0}.

3. Step 2: Multi-class classification

Selected frequent classes from train: 18 available classes in this subset.

The pipeline for every model is `StringIndexer -> VectorAssembler -> StandardScaler -> classifier`, with inverse-frequency `weightCol` from train only.

Validation tuning results:

Model | Best config | Validation macro-F1 | Train time (s)

lr | {'regParam': 0.0, 'elasticNetParam': 0.0} | 0.735 | 20.537

rf | {'numTrees': 40, 'maxDepth': 8} | 0.845 | 5.544

gbt_ovr | {'maxIter': 10, 'maxDepth': 3} | 0.767 | 83.295

Best model: `rf`. Test accuracy=0.958, weighted-F1=0.955, macro-F1=0.511.

Assignment 2 baselines are included in `reports/a3_step2_multiclass.json` for comparison.

Top RF features include: ndvi_p10, mean_b11, mean_b12, brightness, mean_b03, ndwi, mean_b08, mean_b8a, ndvi_p90, mean_b06.

Top-k retraining summary:

k | Test accuracy | Macro-F1 | Weighted-F1

5 | 0.793 | 0.439 | 0.861

10 | 0.815 | 0.443 | 0.881

4. Step 3: Multi-label classification

All 19 CORINE labels are encoded as binary columns. One Logistic Regression binary classifier is trained per class using one train-fitted feature-preparation PipelineModel. Thresholds are selected on validation only.

Test micro-F1=0.627; macro-F1=0.397.

Marine waters and Pastures are strongest; rare or visually mixed classes such as Coastal wetlands, Beaches/dunes/sands, and Industrial/commercial units are weak from patch-level summaries.

5. Step 4: Unsupervised analysis

PCA explained variance for PC1/PC2: [0.7208, 0.1228].

PCA plot: `C:\Users\DELL\Documents\Bigdata-Lab\Lab2-Bigdata\reports\figures\a3_pca_scatter.png`.

Best KMeans k by silhouette: 3 with silhouette=0.789.

The contingency table shows one cluster dominated by Marine waters and others mostly vegetation/agriculture; clusters do not perfectly match land-cover classes because labels are multi-label and features lose spatial texture.

6. Step 5: Scalability and deployment

Scaling experiments used stratified `sampleBy`, two partition counts, and caching. Plot: `reports/figures/a3_scalability.png`.

Train fraction | Partitions | Train rows | Time (s) | Test macro-F1

0.25 | 4 | 1567 | 8.763 | 0.507

0.25 | 8 | 1567 | 5.01 | 0.515

0.5 | 4 | 3240 | 5.793 | 0.547

0.5 | 8 | 3240 | 5.261 | 0.538

1.0 | 4 | 6567 | 4.139 | 0.626

1.0 | 8 | 6567 | 5.42 | 0.585

Model saved to `file:///C:/Users/DELL/Documents/Bigdata-Lab/Lab2-Bigdata/hdfs_mirror/bigearthnet/models/a3_best_pipeline_model` and reloaded metrics matched exactly: `True`.

Batch predictions saved to `file:///C:/Users/DELL/Documents/Bigdata-Lab/Lab2-Bigdata/hdfs_mirror/bigearthnet/predictions` partitioned by country.

Held-out country `Lithuania` accuracy=0.955, macro-F1=0.513.

7. Reproducibility commands

```
.\.venv\Scripts\python.exe scripts\05_a3_features_v2.py --partitions 16
.\.venv\Scripts\python.exe scripts\06_a3_multiclass.py
.\.venv\Scripts\python.exe scripts\07_a3_multilabel.py
.\.venv\Scripts\python.exe scripts\08_a3_unsupervised.py
.\.venv\Scripts\python.exe scripts\09_a3_scalability_deploy.py
```

8. Limitation / deviation notes

• The provided partial dataset contains 18 of 19 class labels and the single-label top-class task has fewer than 35 usable exact-one frequent classes. The scripts request 35 but transparently report the available classes/counts.

• Windows local Spark required local Hadoop native helpers (`winutils.exe`, `hadoop.dll`) for Parquet writes.

Appendix: requirement checkpoint audit

■# Assignment 3 Requirement Checkpoint Audit

Checkpoint created before work: Git branch `checkpoint-before-assignment-3` from commit `d2cf72e`.

Working branch: `assignment-3-work`.

Requirement | Evidence | Status

|---|---|---|

Delete old Assignment 2 plan/spec/audit markdown files | Removed old files under `docs/superpowers/plans`, `docs/superpowers/specs`, `reports/REPORT_OUTLINE.md`, `reports/REQUIREMENT_AUDIT.md` | DONE

Step 1 richer 20m features | `scripts/05_a3_features_v2.py`, `reports/a3_step1_features_v2_validation.json` | DONE

50,000+ patches and 3+ countries | Step 1: 52,607 rows, 4 countries | DONE

Official split, no random split | All scripts use existing `split` column; no `randomSplit`, no CrossValidator/TrainValidationSplit | DONE

Train-only preprocessing inside Pipeline | Step 2 and Step 5 use Pipeline; Step 3/4 prep pipelines fit on train only | DONE

Step 2 exact-one top frequent class task | `scripts/06_a3_multiclass.py`, counts in `reports/a3_step2_multiclass.json` | DONE

Step 2 LR/RF/GBT-OneVsRest grids tuned on validation | `reports/a3_step2_multiclass.json` includes all grid rows and timing | DONE

Step 2 test metrics/confusion/per-class metrics | `reports/a3_step2_multiclass.json` | DONE

Step 2 compare with A2 baselines | `assignment2_baselines` section in `reports/a3_step2_multiclass.json` | DONE

Step 2 RF importances/LR coefficients/top-k retrain | `interpretation` and `top_k_retrain` in Step 2 JSON | DONE

Step 3 19 binary label columns | `scripts/07_a3_multilabel.py`, `reports/a3_step3_multilabel.json` | DONE

Step 3 binary relevance loop + threshold tuning | `reports/a3_step3_multilabel.json` per-class threshold grid | DONE

Step 3 test micro/macro/per-class F1 | `summary` and `per_class` in Step 3 JSON | DONE

Step 4 PCA train pipeline + plot | `reports/a3_step4_unsupervised.json`, `reports/figures/a3_pca_scatter.png` | DONE

Step 4 KMeans k=3..10 silhouette + contingency | `reports/a3_step4_unsupervised.json` | DONE

Step 5 scaling 25/50/100 and two partition counts | `reports/a3_step5_scalability_deployment.json` | DONE

Step 5 model persistence reload exact metrics | `reload_reproduces_test_metrics_exactly: true` | DONE

Step 5 batch inference held-out country partitioned by country | `hdfs_mirror/bigearthnet/predictions/country=Lithuania`, metrics in Step 5 JSON | DONE

Deliverable scripts/config/requirements/report | `scripts/05..09_a3_*.py`, `scripts/a3_config.py`, `requirements.txt`, `reports/a3_final_report.md` | DONE

Known transparent limitations:

• Partial dataset has 18 of 19 labels and the exact-one single-label subset does not contain 35 usable train classes; scripts request 35 and report available classes/counts.

• Local Windows Spark required Hadoop native helper files in `.hadoop/bin`; `.hadoop/` is ignored from Git.